ECON 0100 | Exercise E2 | Competition

Solutions

Answers and work are shown in the red blocks.

Use the animation's competitive firm: $TC = 1000 + q^2$ and $MC = 2q$. Costs include opportunity costs. Answer alone, then compare with a neighbor.

Q1 | Choosing Output

a) The market price is 90. Find total revenue from selling 20 units and from selling 21 units. What is marginal revenue? Explain why it equals the price.

b) At that price, how much should the firm produce? Explain why it should increase output from 40 units but decrease output from 50 units.

c) Calculate the firm's profit at its chosen quantity.

a) Revenue is 1800 at 20 units and 1890 at 21; marginal revenue is 90. **b)** Produce 45 units. **c)** Economic profit is 1025.

a) $TR = Pq$, so $TR(20) = 90(20) = 1800$ and $TR(21) = 90(21) = 1890$. The additional unit adds $1890 - 1800 = 90$ to revenue. The firm takes the market price as given: selling one more unit does not lower the price received on its other units.

b) At 40 units, $MC = 80 < MR = 90$, so increasing output raises profit. At 50 units, $MC = 100 > MR = 90$, so decreasing output raises profit. On rising MC, the best positive output satisfies $2q = 90$, giving $q = 45$.

c) $TR = 90(45) = 4050$ and $TC = 1000 + 45^2 = 3025$. Profit is $4050 - 3025 = 1025$. It is positive, so operating is better than shutting down. Equivalently, $ATC = 3025/45$ and profit is $45(90 - 3025/45) = 1025$.

Q2 | Operating or Leaving

At a price of 40, the best positive output is 20 units. Fixed cost is unavoidable this period but avoidable in future periods by leaving the industry.

a) Calculate profit from operating and from shutting down this period. Which should the firm choose?

b) If this price persists, should the firm stay in the industry in the long run? Explain why this answer can differ from (a).

c) Return to the profitable price of 90 in Q1. Would firms enter or exit? What happens to market supply and price?

a) Operate: lose 600 instead of 1000. **b)** Exit in the long run if the price persists. **c)** At the profitable price of 90, firms enter, supply increases, and price falls.

a) At $q = 20$, revenue is $40(20) = 800$, variable cost is $20^2 = 400$, and total cost is $1000 + 400 = 1400$. Operating profit is $800 - 1400 = -600$. With zero output, revenue and variable cost are zero but fixed cost is still 1000, so shutdown profit is -1000. Operating contributes 400 toward that unavoidable cost. Price is above $AVC = 20$, even though it is below $ATC = 70$.

b) A continuing price of 40 does not cover all economic costs. The firm exits when it can avoid future fixed costs. A short-run loss does not by itself imply shutdown; a persistent economic loss does motivate long-run exit.

c) The profit of 1025 attracts entry. More firms increase market supply and lower price, reducing the incumbent firm's profit. With identical firms, free entry and exit, and unchanged input prices, entry stops at zero economic profit: $P = MC = \min ATC$. Zero economic profit includes compensation for the owner's opportunity costs.

Q3 | A Higher Fixed Cost

Start again, with the market in long-run equilibrium and every firm earning zero economic profit. Firms have the original costs above. Firms are identical, entry and exit are free, and market demand slopes down.

Every firm must now pay an additional fixed expense of 600 each period, regardless of its output. This period's payment is unavoidable; future payments can be avoided by exiting. Demand and input prices do not change.

a) On the graphs, show the initial equilibrium and the immediate effect of the expense on costs. What happens immediately to price, each firm's output, and profit?

b) Show the adjustment as firms enter or exit. Compared with before the expense, what happens in the long run to price, each remaining firm's output, total market output, and the number of firms? What stops the adjustment?

The Market



One Firm



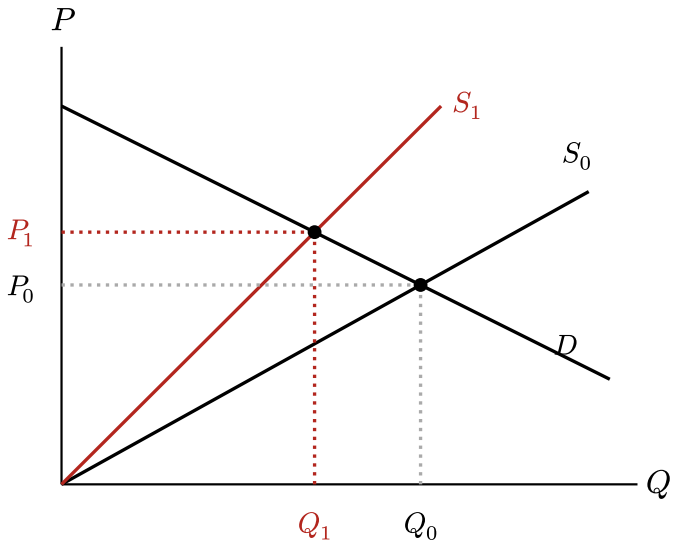
c) Explain how fewer firms can each produce more while the market produces less in total.

a) ATC rises; MC and AVC do not change. Initially price and output are unchanged, and each firm loses 600. **b)** Firms exit. Price and output per survivor rise; total output and the number of firms fall. Exit stops at zero economic profit. **c)** Fewer firms more than offset higher output per survivor.

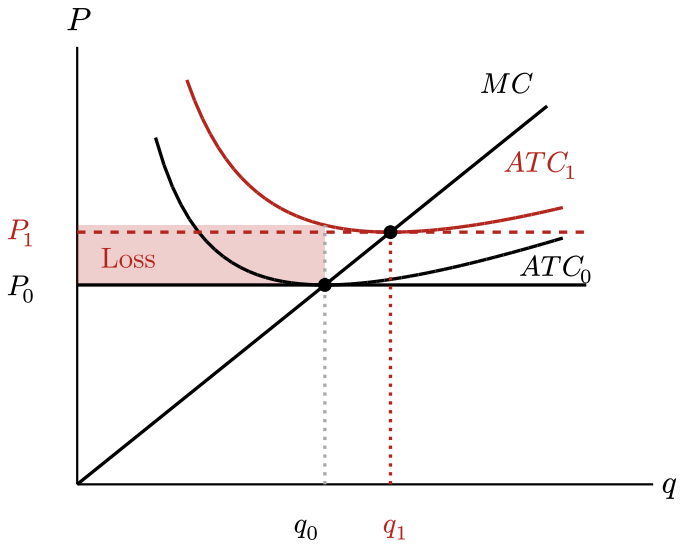
Immediately. ATC rises from $q + 1000/q$ to $q + 1600/q$: the gap is $600/q$, not a parallel shift. $MC = 2q$ and $AVC = q$ are unchanged. Since this period's expense is unavoidable, the same firms keep operating at their original outputs. Market supply and demand are unchanged, so price stays fixed. Each firm's cost rises by 600 with no change in revenue.

After exit. Recurring losses induce exit once future fixed costs can be avoided. Market supply shifts inward. The higher price leads survivors to produce more along their unchanged MC, while buyers purchase less along demand. Exit stops at the new minimum ATC, where economic profit is zero again.

Market After Exit



Firm Before and After



Black marks show the initial equilibrium; red marks show the equilibrium after exit. The shaded rectangle is the immediate loss. S_0 stays fixed immediately after the expense; S_1 reflects fewer firms after exit.

c) $Q = Nq$. Total quantity falls as price rises, even though each survivor produces more. The decline in the number of firms more than offsets the increase in output per firm.